Calculating expectation

Expectation can be calculated. This is done by expressing the likelihood of something happening as a fraction, and then multiplying the fraction by the number of times the occurrence has the chance to happen. This example shows how expectation can be calculated in a game where balls are pulled from a bucket, with numbers ending in 0 or 5 winning a prize.

◁ **Numbered balls** There are 30 balls in the game and 5 are removed at random. The balls are then checked for winning numbers—numbers that end in 0 or 5.

There are 6 winning balls that can be picked out of the total of 30 balls.

number of winning balls in game

6

The total number of balls that can be picked in the game is 30.

total number of balls in game

30

The probability of a winning ball being picked is 6 (balls) out of 30 (balls). This can be written as the fraction 6⁄30, and is then reduced to 1⁄5. The chance of picking a winning ball is 1 in 5, so the chance of winning a prize is 1 in 5.

both parts of the fraction can divide by 6, so it can be reduced

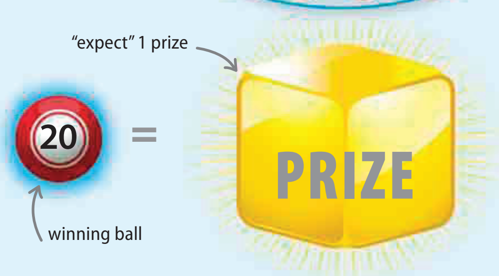
chances of winning ball being picked

$$\frac{6}{30} = \frac{\overset{6 \div 6 = 1}{1}}{\underset{30 \div 6 = 5}{5}}$$

It is expected that a prize will be won exactly 1 out of 5 times. The probability of winning a prize is therefore 1⁄5 of 5, which is 1.

1 prize "probably" won

$$\underbrace{\frac{1}{5}}_{\text{probability of picking a winning ball is 1 in 5}} \times \underbrace{5}_{\text{opportunities to pick a ball}} = 1$$



Expectation suggests that 1 prize will be won if 5 balls are picked. However, in reality no prize or even 5 prizes may be won.

1 prize won?